Professional Quantum Lab Documentation Standards

Student Lab Handout

Session Overview

Objective: Establish professional standards for reproducible quantum lab reports. **Directory**

Structure:

```
/work/pi_phys4340l_uml_edu/student_work/YOUR_USERNAME/
+-- lab1/
|   +-- lab1_final.ipynb
|   +-- screenshots/
|   +-- data/
+-- lab2/
|   +-- lab2_final.ipynb
+-- lab3/
    +-- lab3_final.ipynb
```

Create your lab directories:

```
cd /work/pi_phys4340l_uml_edu/student_work/YOUR_USERNAME
mkdir -p lab{1..4}/{screenshots,data}
```

1 Lab Report Template (MANDATORY 4-Section Format)

Lab X: [Your Experiment Title]

1. Introduction (1-2 sentences REQUIRED)

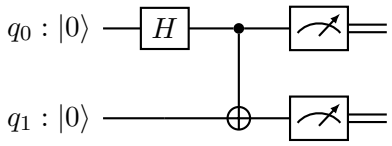
This experiment demonstrates **[CONCEPT]** using **[BACKEND]**. Expected outcome: **[PREDICTION]**.

Example: This experiment demonstrates Bell-state entanglement using `ibm_nairobi`. Expected outcome: 50/50 distribution of $|00\rangle$ and $|11\rangle$ outcomes.

2. Methods (100% Reproducible - MANDATORY)

Backend Details	
Backend	`ibm_nairobi`
Shots	1024
Job ID	`abc123def`
Date	2026-02-13 10:00 EST
Device	Simulator/QPU

Quantum Circuit Diagram:



QASM Code:

```
OPENQASM 2.0;
include "qelib1.inc";
qreg q[2]; creg c[2];
h q[0];
cx q[0], q[1];
measure q[0] -> c[0];
measure q[1] -> c[1];
```

3. Results

IBMQ Histogram Screenshot:

[Insert IBMQ histogram screenshot here]

Raw Results Table:

State	Counts	Probability
00	512	50.0%
11	512	50.0%
Total	1024	100.0%

Python Reproduced Plot (from CSV):

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("data/job_abc123.csv")
plt.figure(figsize=(8,5))
plt.bar(df["State"], df["Counts"], color=["steelblue","coral"], alpha=0.85)
plt.title("Quantum Results (Matches IBMQ?)")
plt.ylabel("Shot Counts")
plt.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.savefig("screenshots/reproduced_results.png", dpi=150)
plt.show()
print("Plot saved: screenshots/reproduced_results.png")
```

4. Discussion (2-3 sentences REQUIRED)

1. Results match theory within ±2% error (1024 shots).
2. The **Job ID abc123def** ensures reproducibility.
3. Deviations arise from readout error or backend noise.

2 Cluster Workflow

1. Navigate to your lab folder:

```
cd /work/pi_phys4340l_uml_edu/student_work/YOUR_USERNAME/lab4
```

2. Launch Jupyter:

```
jupyter notebook
```

3. Save under standardized version names:

Draft	`lab4_analysis_v1.ipynb`
Refined	`lab4_analysis_v2.ipynb`
Graded	`lab4_analysis_final.ipynb`

Required files:

- `lab4_analysis_final.ipynb`
- `screenshots/job_histogram.png`
- `data/job_abc123.csv`

3 Grading Rubric

Criterion	Weight	A-Level	C-Level
Reproducibility	40%	Full run + Job ID + CSV	Missing Job ID
Documentation	30%	All four sections complete	Missing methods
Visualization	20%	Clear, labeled plots	No axis labels
Insight	10%	Quantitative explanation	"It worked" only

ASSIGNMENT: M2 Rewrite

Due: March 2
Path: `lab4/labX_m2_rewrite_final.ipynb`
Requirements:

1. Use the same 4-section template.
2. Include Week X Job ID + screenshot + CSV in `lab4/`.
3. Notebook must execute cleanly from start to finish.

Avoid This Chaos!

```
/student_work/YOUR_USERNAME/     DISASTER
+-- Untitled.ipynb
+-- notebook(2).ipynb
+-- myfile_copy.ipynb
`-- plot.png (which plot?!?)
```

Organize under labX/ directories

Success Mantra

1. **labX/labX_final.ipynb - Never Untitled.ipynb**
2. **"Job IDs or it didn't happen"**
3. **Save all `.csv` and screenshots**
4. **One lab = one directory.**

Quantum is probabilistic. Documentation is your proof.